MIL-G-13672D (MU)
24 October 1968
SUPERSEDING
MIL-G-13672C (MU)
2 August 1965

MILITARY SPECIFICATION

GRENADE, HAND, FRAGMENTATION, DELAY, M56 LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of delay action, fragmentation hand grenade designated as the M56.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1330

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DRAWINGS

U. S. ARMY MUNITIONS COMMAND

7548339 - Container, Fiber, Ammunition M289A2 for Grenade, Hand, Fragmentation, Delay, M56, and Grenade, Hand, Practice, Delay, XM62
7548340 - Box, Packing, Ammunition for Grenade, Hand, Fragmentation, or Grenade, Hand, Practice in Fiber Container M289A2
8840255 - Grenade, Hand, Fragmentation, Delay, M56

PUBLICATIONS

U. S. ARMY MUNITIONS COMMAND

ET - 8840255 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.- Materials shall be in accordance with applicable drawings and specifications.

3.2 Grenade.- The grenade shall comply with all requirements specified on Drawing (dwg.) 8840255 and with all requirements specified in applicable specifications.

3.3 Explosive clearance for crimping.- The crimping flange shall be free of explosives and other foreign material prior to crimping the closing cap. If a chemical cleaner is required, no material other than acetone shall be used. The acetone shall not come in contact with the charge. The test shall be as specified in 4.4.1.

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3.4 Functioning.- The grenade shall not function prematurely (less than three (3) seconds delay) and the detonation when it occurs shall be complete without evidence of pieces of bursting charge or large fragments of spiral coil remaining when tested as specified in 4.4.2.

3.5 Workmanship.- All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined. All packing components shall be dry.

3.6 First article inspection.- This specification makes provision for first article inspection. Requirements for the submission of first article samples shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.- At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

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a. A statement that the lot complies with all quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Statements of findings (certificates of analysis) on the bursting charge and certificates of conformance on all remaining materials when such material is covered by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection

4.2.1 Submission.- The contractor shall submit a sample of thirty (30) completely loaded and packed grenade assemblies to the Government (see 6.1c) for testing in accordance with the provisions of 4.2.2. All parts and materials shall be the same as will be used in production and shall be obtained from the same source of supply. Prior to submission, the sample shall be inspected by the contractor to the degree necessary to assure that it conforms to the requirements of the contract.

4.2.2 Inspections to be performed.- The grenade assemblies shall be tested as specified in 4.4.2 for compliance with the 3.4 requirements.

4.2.3 Rejection.- If any assembly fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon the failure of an assembly to comply with any of the stated requirements.

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4.3 Inspection provisions

4.3.1 Lot formation.- The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. Changes to the process, specifications, or drawings not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105. Unless otherwise approved by the contracting officer the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots of components or sub-assemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot of loaded hand grenades shall contain:

a. Metal parts of one lot interfix number from one manufacturer.

b. Composition B from not more than one lot.

c. Fuzes from not more than two (2) serial lots from one lot interfix number from one manufacturer. These lots shall be consecutively produced lots, but if the adjacent fuze serial lot is not available, the second fuze lot shall not vary from the first fuze lot by more than three (3) serial lot numbers.

4.3.2 Examination.- Inspection for Critical defects (and Major defects, when so specified) shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans

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in accordance with MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect, except where 100 percent inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Loading assembly, prior to assembling closing cap assembly (see dwg. 8838010 covering a detail of dwg. 8840255).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Foreign material on interior....	Visual	01001
Minor:	AQL 1.00 percent		
201.	Evidence of poor workmanship (see 3.5).....	Visual	01002
202.	Disc missing.....	Visual	01004

4.3.2.2 Assembly (see dwg. 8840255).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:			
1.	Safety pin prongs cracked or incorrectly formed.....	Visual	02001
Major:	AQL 1.00 percent		
101.	Fuze loose (can be moved by application of 32 inch pounds torque in tightening direction).	Manual	02003
102.	Fuze of incorrect designation...	Visual	02004
103.	Closing cap assembly crimp not full 360 degrees.....	Visual	02005

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104.	Closing cap assembly loose.....	Manual	02006
105.	Coating compound missing.....	Visual	02007
106.	Fuze not seated or gasket missing.....	Visual	02008
Minor:	AQL 1.50 percent		
201.	Protective coating missing or contains bare spot exposing base material.....	Visual	02010
202.	Marking incorrect, incomplete or illegible.....	Visual	02011
203.	Evidence of poor workmanship (see 3.5).....	Visual	02012

4.3.2.3 Sealed fiber container (see dwg. 7548339).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Container cut or damaged through all impregnated layers.....	Visual	03001
102.	Tape incomplete or badly wrinkled.....	Visual	03002
103.	Metal ends loose, distorted or punctured.....	Visual	03003
Minor:	AQL 1.50 percent		
201.	Tear tab length.....	Scale	03004
202.	Contents loose.....	Manual	03005
203.	Excessive gap between cover and body.....	Visual	03006
204.	Marking incorrect, incomplete or illegible.....	Visual	03007

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4.3.2.4 Sealed packing box (see dwg. 7548340).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Hardware missing, broken or loose	Visual	04001
102.	Strapping missing, broken or loose	Visual	04002
103.	Contents loose	Manual	04003
104.	Box damaged	Visual	04004
105.	DOD symbol misleading or unidentifiable	Visual	04005
Minor:	AQL 1.50 percent		
201.	Strapping improperly assembled	Visual	04006
202.	Metallic seal unsealed or improperly positioned	Visual	04007
203.	Handle insecure	Manual	04008
204.	Marking incorrect, incomplete or illegible	Visual	04009

4.3.3 Production testing

4.3.3.1 Explosive clearance for crimping (see 3.3) - Major defect - Code No. 01003.- This inspection shall be performed 100 percent. The method of inspection shall be by "Visual" examination prior to crimping the closing cap. Any assembly which fails to comply with the specified requirement shall be classed defective and removed from the lot. The inspection shall be performed as specified in 4.4.1.

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4.3.3.2 Functioning (see 3.4 and Table I)

TABLE I

Defect	Classification	Code No.
Premature (see 3.4)	Critical	02013
Failure to function (dud) (see 3.4)	Major	02014
Detonation incomplete (see 3.4)	Major	02015

Ten (10) grenade assemblies shall be selected from each lot for test as specified in 4.4.2 using equipment in accordance with 4.3.4. If a critical defect occurs or if two (2) or more major defectives are found, the lot shall be rejected. If only one major defective is found, a second sample of twenty (20) assemblies shall be selected and tested. If a critical defect occurs in the second sample or if the combined number of major defectives in the first and second sample equals or exceeds two (2), the lot shall be rejected.

4.3.4 Inspection equipment.- Equipment Tabulation Number ET - 8840255 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 Government rights to documentation.- Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance.- Supply and maintenance of the equipment listed on the ET shall be in accordance with MIL-I-45607.

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4.4 Test methods and procedures

4.4.1 Explosive clearance for crimping.- This test shall be performed by visual inspection of the required area for presence of explosive or other foreign material.

4.4.2 Functioning.- The grenade assembly shall be mounted in the specified test fixture in a suitable barricaded arena capable of withstanding the detonation effects. The safety pin shall be removed by remote control and the grenade observed for compliance with the requirements.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5.1.1 Level A.- The grenade assembly shall be packaged in accordance with dwg. 7548339.

5.2 Packing

5.2.1 Level A.- The packaged grenade assemblies shall be packed in accordance with dwg. 7548340.

5.3 Marking.- The container and boxes shall be marked as specified on dwgs. 7548339 and 7548340.

6. NOTES

6.1 Ordering data.- Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards.- Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

c. Submission of first article quantity.- Instructions as to the location of the facility at which the evaluation of the first article quantity shall be carried out may be obtained from the contracting officer.

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6.2 Inspection code numbers.- The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.- The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists (EL).- Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.- The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.- Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, design alternate inspection equipment or use comparable commercial

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equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.- Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.- All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND1. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

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6.4 Submission of loaded test data and ammunition data cards.- In addition to the normal distribution of records, one copy of all loaded test data and ammunition data cards shall be forwarded to each of the following:

Commanding Officer
Picatinny Arsenal
Dover, New Jersey 07801
ATTN: (1) SMUPA-ND5
(2) SMUPA-DC.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1330-A-125

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